Characteristics of vacuum diodes and Franck-Hertz experiment

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We measured the characteristic line of a vacuum diode and confirmed the validity of the Schottky-Langmuir law and the Richardson equation in their range of application. The work function of wolfram was determined to $W_\mathrm{A} = (3.399 \pm 0.031)\,\mathrm{eV}$. Using the Franck-Hertz experiment we dertermined the energy of optical transitions in neon and mercury atoms. In neon the energetic distance between the ground state and the 3p orbital is $E_\mathrm{main}^\mathrm{exp} = (19.425 \pm 0.50)\,\mathrm{eV}$ and the distance between the 3p and the $3\mathrm{s}\,^3\mathrm{P}_2$ orbital is $E_\mathrm{sec}^{exp} = (2.33 \pm 0.71)\,\mathrm{eV}$. For the mercury atomm the energy diference between the ground state and the $6\mathrm{s}^1 6\mathrm{p}^1$ orbital is $E_\mathrm{exp} = (4.84 \pm 0.13)\,\mathrm{eV}$.

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1 Introduction

Vacuum diodes are electric control elements which are rectifiers. They are important parts of sound systems. In the following experiment we will examine the anode characteristic curve and observe thermionic emission and shielding due to space charge.
In the Franck-Hertz experiment a more complex diode filled with gas is used. With this experiment one could confirm the discret energy levels of atoms. The process is based on inelastic scattering of electrons with the gas atoms. We will look at neon and mercury as filling gas.

2 Vacuum diodes

2.1 Setup and theoretical background

A vacuum diode consists of a vacuum tube with a heated cathode and an anode inside the tube. The cathode is heated by a heating voltage U_H. Due to the thermionic emission, the cathode emits electrons. By applying a voltange U_A between anode and cathode, the emitted electrons get accelerated towards the positivly charged anode. One can measure a current I_A at the anode. If the voltage U_A is applied in the opposing direction, one can not measure a current I_A, because the anode is negativly charged. Due to this the vacuum diode is a rectifier.

When looking at high anode voltages U_A, the current I_A is limited by the number of emitted electrons. This number depends on the heating voltage U_H and with that it depends on the temperature T. The current density j in that case follows the Richardson equation (1)

$$j = CT^2 \mathrm{e}^{-\frac{W_\mathrm{A}}{k_\mathrm{B}T}} \tag{1}$$

where C is a material constant, $k_\mathrm{B} = 1.380649 \cdot 10^{-23}\,\frac{\mathrm{J}}{\mathrm{K}}$ [2] is Boltzmann´s constant and W_A is the work function of the cathode material. One can derive this equation from the Fermi-Dirac statistics of the electron energy. This statistic devolves into the Boltzmann statistics for high work functions or low temperatures.

In the case of low anode voltages, the cloud of emitted electrons near the cathode shields the electric field due to its space charge. This leads to a lower current because not all emitted electrons can contribute to the current. In this case, the current density of the vacuum diode follows the Schottky-Langmuir space charge law (2)

$$j = \frac{4}{9}\epsilon_0 \sqrt{\frac{2e}{m_\mathrm{e}}} \frac{(U_\mathrm{A} - U_\mathrm{C})^{\frac{3}{2}}}{d^2} \tag{2}$$

where $\epsilon_0 = 8.8541878128 \cdot 10^{-12}\,\frac{\mathrm{F}}{\mathrm{m}}$ is the electric field constant, $e = 1.602176634 \cdot 10^{-19}\mathrm{C}$ the elementary charge, $m_\mathrm{e} = 9.1093837015 \cdot 10^{-31}\mathrm{kg}$ the electron mass [2], d the distance between anode and cathode and U_C is the contact voltage.

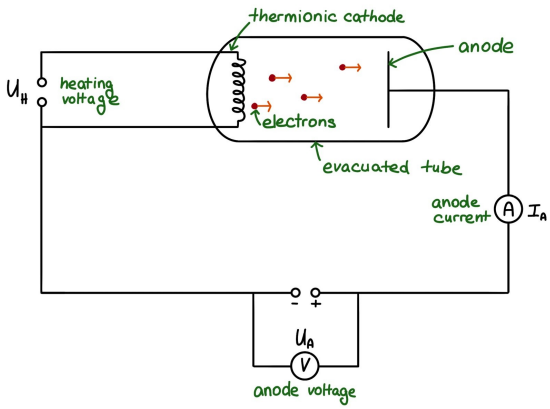

Fig. 1: Setup of a vacuum diode. Increasing the anode voltage U_A leads to an increasing anode current U_A because the electric force on the electrons increases. If the voltage has the opposite polarity, there is no current due to an opposing electric field.

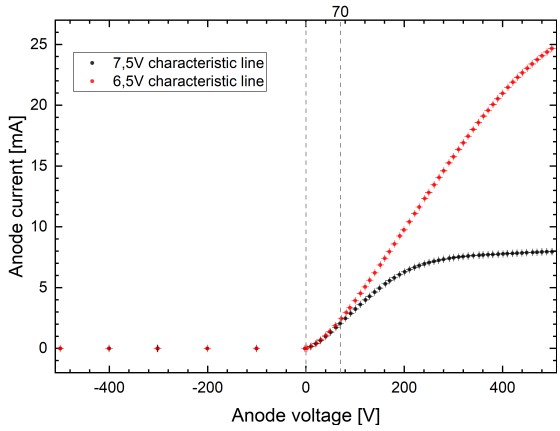

Fig. 2: Characteristic lines of a vacuum diode at two different heating voltages U_H. If the applied voltage U_A is negative, no anode current I_A flows. Once U_A is positiv, I_A follows the Schottky-Lagmuir space charge law (2). If the voltage is high enough, the course of the characteristic curve enters the saturation range and the current flattens out. For $U_H = 7.5\,V$ the applied voltage is not sufficient to reach this range.

The contact voltage is caused by the difference in work function when two different materials meet. This produces a separation of charge, which causes the contact voltage. For the vacuum diode the contact voltage is the voltage that has to be applied to measure no current.

2.2 Characteristic line of a vacuum diode

The anode voltage U_A is increased step by step for two different heating voltages $U_H = 6.5\,V$ and $U_H = 7.5\,V$. The anode current I_A and voltage U_A are measured with two multimeters *Agilent* 33220A and the resulting characteristic curves are shown in Fig. 2. For large negative voltages no anode current flows, because the thermal energy of the emitted electrons is not sufficient to counteract the anode voltage. Once the anode voltage approaches zero and then increases from there, the anode current follows Schottky-Langmuir space charge law (2). From about $U_A = 70\,V$, the space charge is increasingly sucked off by the anode voltage, which further reduces the shielding effect. As a result, the emitted electrons reach the anode more unobstructed. As soon as the anode voltage is high enough for all emitting electrons to be extracted, the characteristic curve enters the saturation range and the current only increases slightly. From this point on, the anode current is described by the Richardson-equation (1), since I_A only depends on how many electrons are emitted by the hot cathode.
For $U_H = 6.5\,V$, the flattening of the characteristic curve for high voltages is clearly visible. If the heating voltage $U_H = 7.5\,V$ is applied, so many electrons are emitted that the high voltage U_A applied is not sufficient to observe the saturation range of the characteristic line.

2.3 Space charge area

In order to examine more precisely whether the current course of the two characteristic curves for low anode voltage follow the Schottky-Langmuir space charge law (2), the characteristic curve between 0 V and 70 V is plotted for both heating voltages. The data is plotted in double-logarithmic form and a fit of the form $y = a\,x^b$ is applied (Fig. 3 and Fig. 4).
In order to check now whether the course corresponds to the Schottky-Langmuir space charge law (2), the exponent b of the fit is compared to the exponent $b_{SL} = \frac{3}{2}$. The resulting values for b are listed in Tab. 1. For both heating voltages the determined exponents within the σ-band deviate from the value $b_{SL} = \frac{3}{2}$. The reason for this is the neglected contact voltage U_C. Since a current flows even with a

U_H [V]	6.5	7.5
b unadjusted	1.403 ± 0.014	1.3491 ± 0.0091
U_C [V]	-3.86 ± 0.53	-2.53 ± 0.17
b adjusted	1.489 ± 0.020	1.4974 ± 0.0075

Tab. 1: Calculated contact voltages U_C for both heating voltages U_H and the unadjusted exponent b as well as the adjusted exponent b.

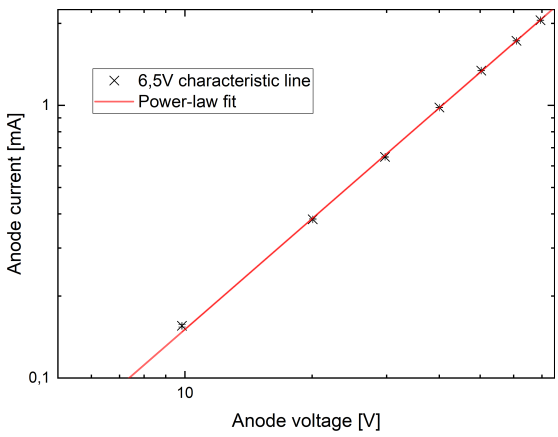

Fig. 3: Section of the characteristic line for $U_\mathrm{H} = 6.5\,\mathrm{V}$ which shows the range of validity of the Schottky-Langmuir space charge law. The data inside the area $U_\mathrm{A} \in [0\,\mathrm{V};\ 70\,\mathrm{V}]$ is plotted in double-logarithmic form and a linear fit is applied.

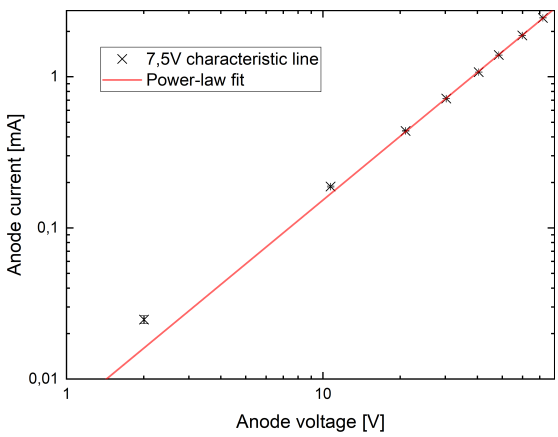

Fig. 4: Section of the characteristic line for $U_\mathrm{H} = 7.5\,\mathrm{V}$ which shows the range of validity of the Schottky-Langmuir space charge law. The data inside the area $U_\mathrm{A} \in [0\,\mathrm{V};\ 70\,\mathrm{V}]$ is plotted in double-logarithmic form and a linear fit is applied.

low negative anode voltage up to a certain voltage U_C. This must be considered for the calculation of b, as can be seen in (2).

In order to determine U_C it is assumed that the meausured characteristic line follows the Schottky-Langmuir space charge law (2) at low voltage U_A. $I_\mathrm{A}^{2/3}$ is plotted against U_A and a linear fit is applied to the data (Fig. 5 and Fig. 6). The contact voltage U_C is obtained from the intersection of the linear fit with the voltage axis. The resulting contact voltages

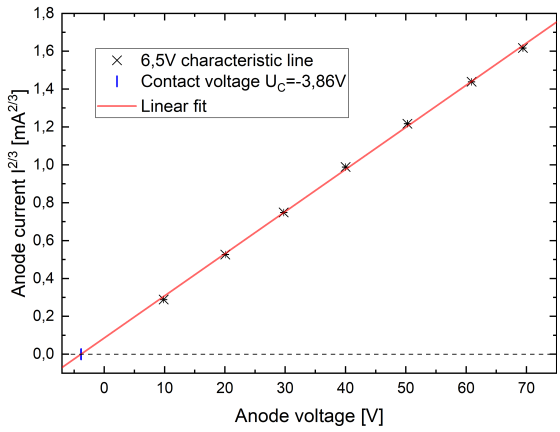

Fig. 5: Determining of the contact voltage U_C for $U_\mathrm{H} = 6.5\,\mathrm{V}$. $I_\mathrm{A}^{2/3}$ is plotted against U_A and a linear fit is applied. The intersection of the linear fit and the voltage axis results in the contact voltage $U_\mathrm{C} = -3.86\,\mathrm{V}$

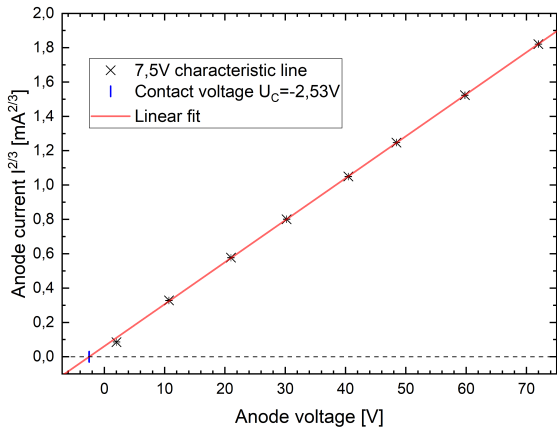

Fig. 6: Determining of the contact voltage U_C for $U_\mathrm{H} = 7.5\,\mathrm{V}$. $I_\mathrm{A}^{2/3}$ is plotted against U_A and a linear fit is applied. The intersection of the linear fit and the voltage axis results in the contact voltage $U_\mathrm{C} = -2.53\,\mathrm{V}$

are shown in Tab. 1.

In order to determine the adjusted exponent b, I_A is no longer plotted against U_A but against $U_\mathrm{A} - U_\mathrm{C}$ in a double-logarithmic form. The fit $y = a\,x^b$ is now applied and the adjusted value of the exponent is de-

termined from it. The resulting values are listed in Tab. 1. For both heating voltages, the two values of b within the σ-band coincide with the theoretical exponent $b_{\mathrm{SL}} = \frac{3}{2}$. As a result, the validation of the Schottky-Langmuir space charge law (2) can be assumed for low voltages.

2.4 Saturation range and Richardson equation

Once the characteristic line reaches the saturation range, the anode current I_{A} doesn't change with increasing the voltage U_{A} since the number of emitted electrons is limited. This number is only dependant on the temperature of the cathode and therefore on the applied heating voltage. The relation between the temperature of the used wolfram cathode and the applied heating voltage is described in the following equation [1]:

$$T(U_{\mathrm{H}}) = 1616\,\mathrm{K} + 97\,\frac{\mathrm{K}}{\mathrm{V}}\,U_{\mathrm{H}} \tag{3}$$

In order to measure the relation between the saturation current and the heating voltage U_{H}, an anode voltage of $U_{\mathrm{A}} = 500\,\mathrm{V}$ is applied and U_{H} is increased from $5\,\mathrm{V}$ to $7\,\mathrm{V}$ in $0.1\,\mathrm{V}$ steps. With the aim of controlling whether the measured data follows the expected theoretical course from the Richardson equation (1), $\frac{I}{T^2}$ is plotted on a logarithmic scale against $\frac{1}{T}$ (Fig. 7). The graphic shows that for increasing temperatures the data points drift away from the exponential fit to smaller values. This is because the applied anode voltage of 500V isn't big enough to reach the saturation range of the characteristic line at such high temperatures. Therefore those data points are not used for the overlaying fit.

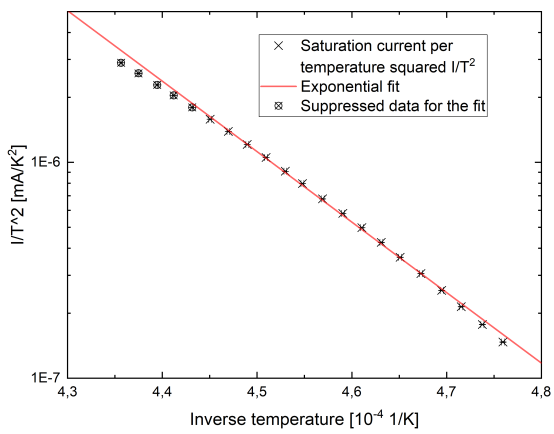

Fig. 7: Relation between the saturation current I_{A} and the temperature T, which is calculated using (3). The suppressed data isn't used for the applied exponential fit.

In order to determine the work function with the exponential fit, (1) is rearranged to:

$$\ln \frac{I_{\mathrm{A}}}{T^2} = a - \frac{W_{\mathrm{A}}}{k_{\mathrm{B}}}\,\frac{1}{T} \tag{4}$$

Using the slope b of the fit and equates it with -$\frac{W_{\mathrm{A}}}{k_{\mathrm{B}}}$, this results in the work function:

$$W_{\mathrm{A}} = -b\,k_{\mathrm{B}} \tag{5}$$

This results in the value $W_{\mathrm{A}} = (3.399 \pm 0.031)\,\mathrm{eV}$. Compared to the literature value of wolfram $W_{\mathrm{A}} \in [4.54\,\mathrm{eV};\ 4.60\,\mathrm{eV}]$ the measured value does not coincide with it [4]. Impurities in the anode material and the age of the used anode could lead to this deviation.

3 Franck-Hertz experiment

3.1 Theoretical background

It is known from quantum physics that atoms can only absorb or emit energy in a quantized form. Atoms therefore have discrete energy levels. Atoms can be excited to higher energy levels for example by absorbing photons with the right energy. This type of excitation leads to discrete absorption spectra of atoms, since only the photons whose energies correspond to the difference of two energy levels are absorbed. Analog atoms also have discrete emission spectra, since an atom emits a photon when it falls to a lower energy level.
Inelastic collisions with electrons are another way to excite an atom. In this case the electron confers a part of its kinetic energy to the atom, where the transfered energy is equivalent to the difference of two energy levels.

3.2 Setup and functionality of the experiment

The experiment is performed using a gas-filled tube which contains a hot cathode on one side and an anode on the other side. In contrast to the vacuum diode, two grids are installed between the cathode and the anode (Fig. 8).

A low suction voltage U_1 is applied between the cathode and the first grid, which causes the electrons, that are emitted from the cathode to move in the direction of the grid. This ensures that the electrons leave the range of validity of the Schottky-Langmuir space charge law and that no shielding effect takes place. A higher voltage U_2 is applied between the two grids, which accelerates the electrons towards the second grid. Due to the passage of the voltage, the electrons receive the kinetic energy $E_{\mathrm{kin}} = e\,U_2$, where e is the elementary charge. As soon as U_2 is large

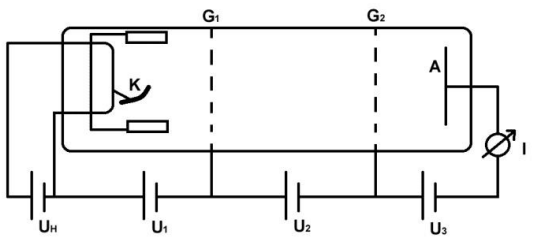

Fig. 8: Schematic setup of the Franck-Hertz experiment. The hot cathode is operated with the voltage U_{H} and the acceleration voltage U_2 is applied between the two grids. U_1 is used as suction voltage and U_3 is used as countervoltage. [3]

enough that the kinetic energy of the electrons is sufficient to energetically excite the gas atoms, they collide inelastic and transfer their energy to the atoms. The excited atoms now fall back into their ground state emitting a photon and a glowing line becomes visible in the tube. If U_2 is further increased, the glowing line in the tube moves in the direction of the cathode because the electrons already have enough energy to interact with the atoms after a shorter distance. If the voltage is high enough, several lines appear in the tube, since the electrons are accelerated again after they have released their energy for the first time and once again have enough energy to excite an atom. In order to be able to measure the drop in the kinetic energy of the electrons, a low countervoltage U_3 is applied between the second grid and the anode. This means that slow electrons that have recently given up their energy do not hit the anode, but are caught by the second grid. These trapped electrons are therefore not measured as a current passing the ammeter behind the anode. If the measured current intensity is plotted as a function of the acceleration voltage, the values of U_2, which correspond to a multiple of the excitation energy of the atoms, show strong drops in the current intensity.

3.3 Franck-Hertz experiment with neon

For the execution of the Franck-Hertz experiment with a tube filled with neon gas, we increase the acceleration voltage U_2 from 0 V to approximately 70 V and measure the anode voltage which is proportional to the current detected at the anode.

The main maxima of the curve in Fig. 9 follows the theoretically expection. In addition to this, the curve also has three smaller secondary maxima. One can explain those secondary maxima with the fact, that there are more than one possible discrete excited state of the neon atoms. There can also be transitions between to exited states. This causes the secondary maxima.

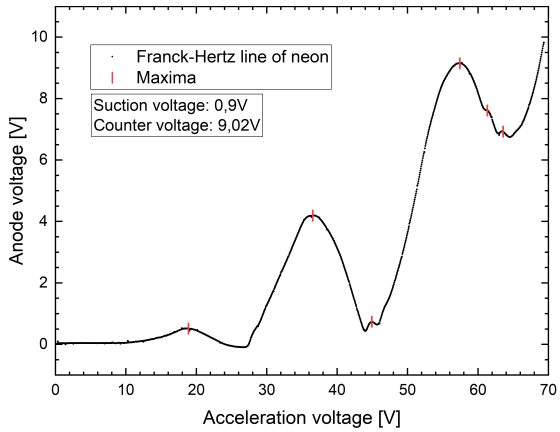

Fig. 9: Franck-Hertz curve of a neon filled tube. One can see three main maxima and three secondary maxima. The maxima are marked in the plot.

Maxima	Energy difference [eV]
$1 \to 2$	18.14 ± 0.71
$2 \to 3$	8.24 ± 0.71
$2 \to 4$	20.71 ± 0.71
$4 \to 5$	3.71 ± 0.71
$4 \to 6$	6.04 ± 0.71
$5 \to 6$	2.33 ± 0.71

Tab. 2: Energy differences between the maxima of the Franck-Hertz curve of neon in Fig. 9.

The energy differences between the maxima in Fig. 9 are shown in Tab. 2. For that the voltage differences are identified and multiplied with the elementary charge e to turn into enrgy differences. The maxima 1, 2 and 4 are the main maxima. The mean value of their energy difference is $E_{\mathrm{main}}^{\mathrm{exp}} = (19.43 \pm 0.50)\,\mathrm{eV}$. The energy difference between two neighbouring secondary maxima is $E_{\mathrm{sec}}^{exp} = (2.33 \pm 0.71)\,\mathrm{eV}$ according to the transition $5 \to 6$. According to the term diagram in Fig. 13 one can correlate the main maxima to a transition from the ground state in the 2p orbital into the 3p orbital. This transition has a energy difference of $E_{\mathrm{main}}^{\mathrm{lit}} = 18.9\,\mathrm{eV}$. The measured value coincides with this theortical value inside its margin of error. The energy difference between the secondary maxima 5 and 6 excactly coincides with the transition between the excited state in the 3p orbital and the excited state in the $3\mathrm{s}\,^3\mathrm{P}_2$ orbital with an energy difference of $E_{\mathrm{sec}}^{\mathrm{lit}} = 2.33\,\mathrm{eV}$. The

lower peak of the secondary maxima compared to the main maxima leads to the conclusion that the transitions responsible for the secondary maxima are much less likely to happen than the transition responsible for the main maxima.

We come to the conclusion that an electron releases its kinetic energy via inelastic scattering to the neon atom that transits from the ground state into the excited state with 3p orbital. The atom relaxes under emission of yellow/orange light into the $3s\,^3P_2$ orbital. Than the atom either relaxes into the ground state or it receives energy via inelastic scattering from an electron to transist back to the excited 3p orbital. The second option creates the secondary maxima. This excitation can happen several times. This explains, why there can be multiple secondary maxima after one main maximum.

If one apply a high enough suction voltage U_1 to get a high electron current between the two grids one can observe areas that emit yellow/orange light. The values of the acceleration voltage U_2 when a new luminous line in the tube appears are listed in Tab. 3. These lines appear at the second grid an move with increasing accelaration voltage towards the first grid. This is caused by the higher electric field, that leads to electrons with enough energy for the inelastic scattering closer at the first grid.

Number of lines	Acceleration voltage U_2 [V]
1	20.5 ± 1
2	39.6 ± 1
3	62.0 ± 1

Tab. 3: Accelleration voltages when a new luminous line appears in the neon filled Franck-Hertz tube. The errors are roughly estimated.

The values of the acceleration voltage is roughly estimated because the lines have a low luminosity when they appear. Due to that it is hard to detect them at the lowest possible voltage. On the level of accuracy the distance between the lines coincides approximately with the energy difference between the main maxima. This supports our conclusion that the emission of the light happens right after the inelastic scattering that causes the main maxima.

3.4 Franck-Hertz experiment with mercury

We execute an analogue experiment with a Franck-Hertz tube filled with gaseous mercury at a temperature of 170 °C. We varied the acceleration voltage from 0 V to approximately 32 V. The measured Franck-Hertz curve is shown in Fig. 10.

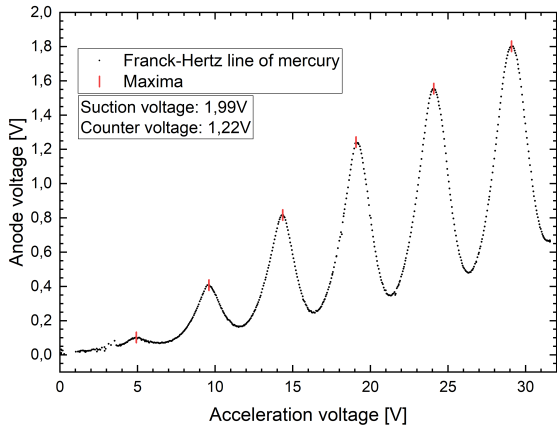

Fig. 10: Franck-Hertz curve of a tube filled with gaseous mercury. One can see six maxima that are marked in the plot.

Compared to the experiment with neon, the curve of mercury has more and sharper maxima. This improves the statistical significance of the data. The energy differences between neighboring maxima are listed in Tab. 4. The mean value of the energetic distance between two neighboring maxima is $E_{\text{exp}} = (4.84 \pm 0.13)$ eV. This value coincides with the transition from the ground state ($6s^2$ orbital) into the exited state with a $6s^1 6p^1$ orbital. The energy difference between these two states is $E_{\text{lit}} = 4.89$ eV according to the term diagram of mercury in Fig. 14. This energy difference corresponds to photons with a wavelength $\lambda = 253.5$ nm. This is in the range of ultraviolet light and is not visible for the human eye.

Maxima	Energy difference [eV]
$1 \to 2$	4.68 ± 0.28
$2 \to 3$	4.75 ± 0.28
$3 \to 4$	4.73 ± 0.28
$4 \to 5$	5.01 ± 0.28
$5 \to 6$	5.01 ± 0.28

Tab. 4: Energy differences between the maxima of the Franck-Hertz curve of mercury in Fig. 10.

One can determine the contact voltage U_C of the Franck-Hertz tube as the voltage where no anode current can be detected. This is the case when the applied electric field compensates the contact electric field that is caused by the different work functions of cathode and anode. The applied electric field is produced by the combination of the suction voltage U_1

and the acceleration voltage U_2. With that an electron gains in this resulting electric field a total energy of

$$E_{\text{tot}} = e(U_1 + U_2 - U_{\text{C}}). \tag{6}$$

In each scattering zone, the electron releases the recently calculated energy $\Delta E = E_{\text{exp}} = (4.84 \pm 0.13)\,\text{eV}$ to an atom. So at the nth maximum the electron has released $E_{\text{tot}} = n \cdot \Delta E$ to the atoms. With that the contact voltage can be calculated with

$$U_{\text{C}} = \frac{n}{e}\,\Delta E - (U_1 + U_2^{\text{n}}) \tag{7}$$

The weighted mean value of the contact voltage of all maxima is $U_{\text{C}} = (-1.98 \pm 0.16)\,\text{V}$. This value is nonzero in its margin of error. Due to that, one can assume that the anode and the cathode are made out of different materials. Contamination of the anode or the cathode could be an other source of the difference in work function. With the executed experiment we can not clarify which of these options causes the contact voltage.

4 Conclusion

We measured the characteristic line of a vacuum diode for two different heating voltages U_{H}. Inside the range of the anode voltage between $-500\,\text{V}$ and $500\,\text{V}$ we identified for $U_{\text{H}} = 6.5\,\text{V}$ the blocking zone for negative voltages, the space charge area for low positive voltages and the saturation range for high positive voltages. In comparison to that the saturation range was not reached for $U_{\text{H}} = 7.5\,\text{V}$. A power law fit applied to the data inside the space charge area $U_{\text{A}} \in [0\,\text{V}; 70\,\text{V}]$ leads to the exponents $b_{6.5\,\text{V}} = 1.403 \pm 0.014$ and $b_{7.5\,\text{V}} = 1.3491 \pm 0.0091$. Those do not coincide with the theoretically expected value of $\frac{3}{2}$. If the data is adjusted for the coantact voltages $U_{\text{C}}^{6.5\,\text{V}} = (-3.86 \pm 0.53)\,\text{V}$ and $U_{\text{C}}^{7.5\,\text{V}} = (-2.53 \pm 0.17)\,\text{V}$ we get the exponents $b_{6.5\,\text{V}}^{\text{adj}} = 1.489 \pm 0.020$ and $b_{7.5\,\text{V}}^{\text{adj}} = 1.4974 \pm 0.0075$. The adjusted exponents coincede with the theoretical expection. We measured the anode current I_{A} for a fixed anode voltage $U_{\text{A}} = 500\,\text{V}$ and heating voltages U_{H} in the range of $[5\,\text{V}; 7\,\text{V}]$ and analyzed its relation to the saturation current. Out of this relation we get the work function of the wolfram cathode $W_{\text{A}} = (3.399 \pm 0.031)\,\text{eV}$. This does not coincide with the literature value.
We excecuted the Franck-Hertz experiment with a neon filled tube. We measured the characteristic line and determined the main and secondary maxima. The mean energy difference between two main maxima is $E_{\text{main}}^{\text{exp}} = (19.425 \pm 0.50)\,\text{eV}$, the energy difference between two secondary maxima is $E_{\text{sec}}^{exp} = (2.33 \pm 0.71)\,\text{eV}$. The main maxima are caused by a transition in the neon atom from the ground state into the exited 3p orbital state with a relative energy of $E_{\text{main}}^{\text{lit}} = 18.9\,\text{eV}$. Then the neon atom relaxes to the $3\text{s}\,^3\text{P}_2$ orbital while emitting a visible yellow/orange photon. Then it can be reexcitated to the 3p orbital, causing the secondary maxima. The energetic distance between those two states is $E_{\text{sec}}^{\text{lit}} = 2.33\,\text{eV}$. We also measured the characteristic line of a mercury filled Franck-Hertz tube and determined its maxima. The mean energy difference between two neighboring maxima is $E_{\text{exp}} = (4.84 \pm 0.13)\,\text{eV}$. This coincides with the energy difference $E_{\text{lit}} = 4.89\,\text{eV}$ of an atomic transition of the mercury atom from the ground state into the $6\text{s}^1 6\text{p}^1$ orbital. The contact voltage between anode and cathode could be determinded to $U_{\text{C}} = (-1.98 \pm 0.16)\,\text{V}$.

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[3] `https://owlcation.com/stem/The-Frank-Hertz-Experiment`, last visited 8th March, 2023

[4] `https://www.fh-muenster.de/ciw/downloads/personal/juestel/juestel/chemie/Austrittsarbeit.pdf`, last visited 23th March, 2023

5 Appendix

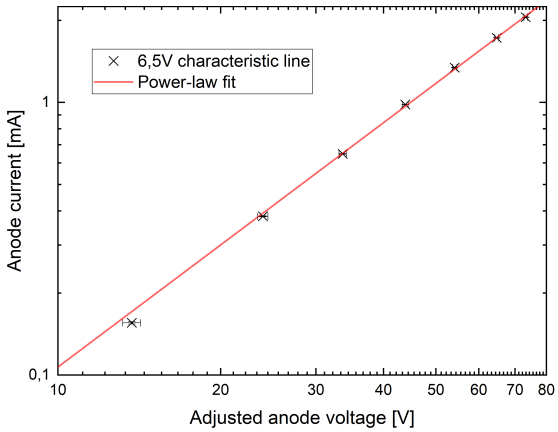

Fig. 11: Section of the characteristic line for $U_\mathrm{H} = 6.5\,\mathrm{V}$ which shows the range of validity of the Schottky-Langmuir space charge law. $U_\mathrm{A} - U_\mathrm{C}$ inside the area $U_\mathrm{A} \in [0\,\mathrm{V};\ 70\,\mathrm{V}]$ is plotted in double-logarithmic form and a linear fit is applied.

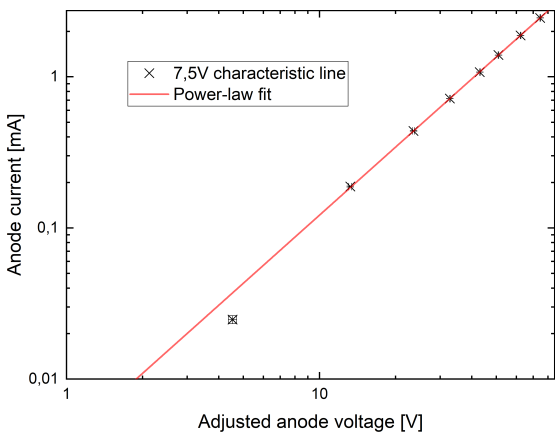

Fig. 12: Section of the characteristic line for $U_\mathrm{H} = 7.5\,\mathrm{V}$ which shows the range of validity of the Schottky-Langmuir space charge law. $U_\mathrm{A} - U_\mathrm{C}$ inside the area $U_\mathrm{A} \in [0\,\mathrm{V};\ 70\,\mathrm{V}]$ is plotted in double-logarithmic form and a linear fit is applied.

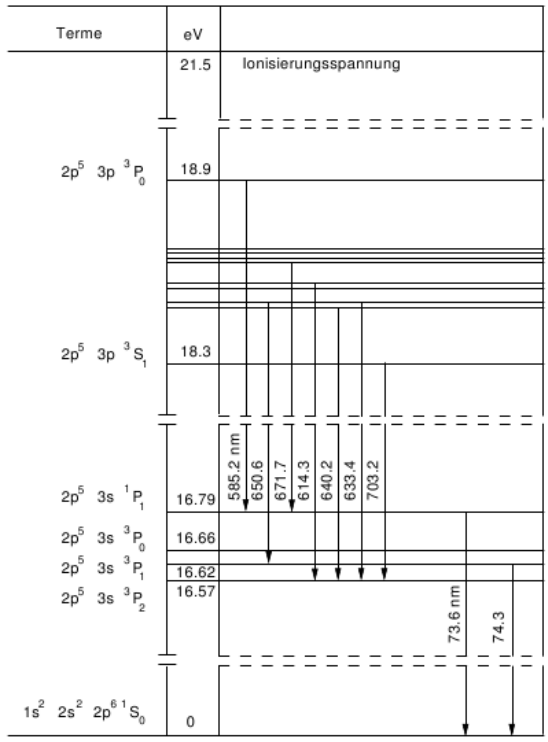

Fig. 13: Term diagram of neon with the allowed transitions and their corresponding energy. [1]

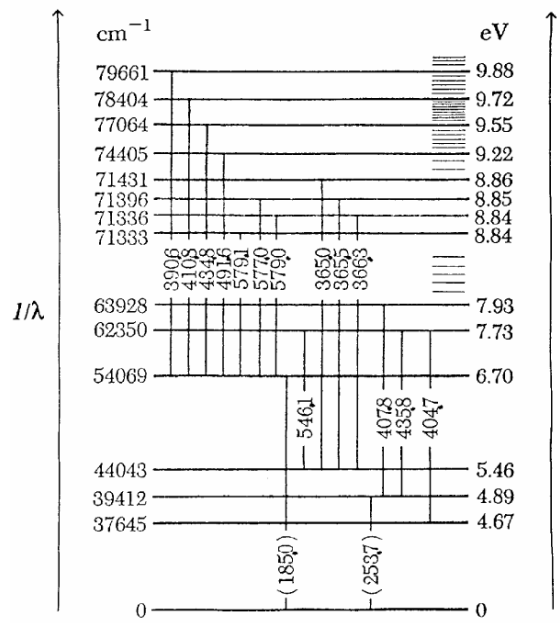

Fig. 14: Term diagram of mercury with the allowed transitions and their corresponding energy. [1]